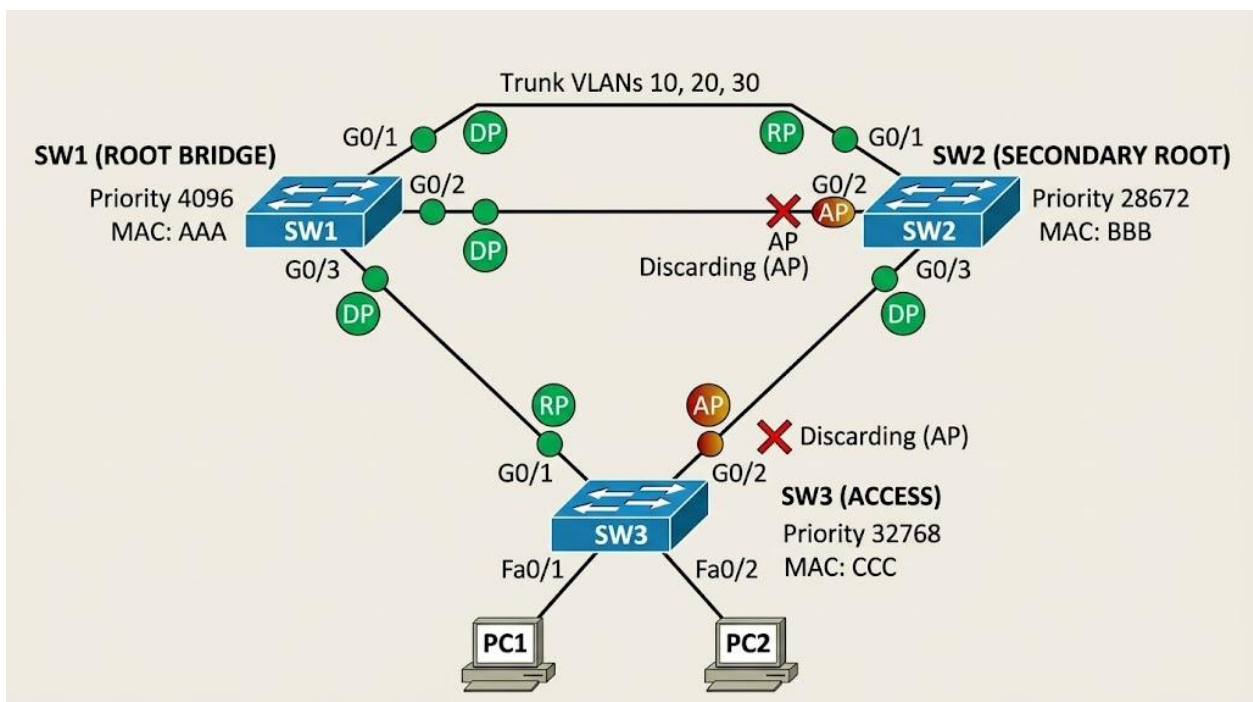


Rapid PVST+ LAB 16

Objectives

- Configure PortFast on end-device/access ports.
- Configure BPDU Guard to protect PortFast access ports.
- Configure Root Guard on designated switch-to-switch ports.
- Configure Loop Guard and understand protection against unidirectional/STP failure conditions.
- Verify STP using show commands and perform controlled troubleshooting tests.



Understand Port Roles and States

Term	Meaning	What to look for
Root Bridge	Switch with the lowest Bridge ID for that VLAN.	All active STP paths point toward it.
Root Port (RP)	Best path from a non-root switch toward the root bridge.	One root port per non-root switch per VLAN.

Designated Port (DP)	Port selected to forward for a network segment.	Normally Forwarding.
Alternate Port	A backup path toward the root; provides loop protection.	Normally Discarding.
PortFast	Rapidly moves an edge/access port to Forwarding.	Use only toward end devices.
Discarding	Does not forward user frames or learn MAC addresses.	Rapid PVST+ state.
Learning	Learns source MAC addresses but does not yet forward user frames.	Rapid PVST+ state.
Forwarding	Forwards frames and learns MAC addresses.	Normal active state.

Part E – Configure Access Ports + PortFast And BPDU Guard

SW3:

configure terminal

```
interface fastEthernet 0/1
  description PC1_VLAN10
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
  spanning-tree bpduguard enable
  no shutdown
```

```
interface fastEthernet 0/2
  description PC2_VLAN20
  switchport mode access
  switchport access vlan 20
  spanning-tree portfast
  spanning-tree bpduguard enable
  no shutdown
```

```
end
```

Recommended global alternative for larger Company:

```
configure terminal
spanning-tree portfast default
spanning-tree portfast bpduguard default
end
```

Important: Do NOT enable PortFast on switch-to-switch trunk links.

BPDU Guard

BPDU Guard protects an edge/PortFast port. If a switch unexpectedly sends BPDUs into a protected access port, Cisco can place the interface into an err-disabled state. This prevents an unauthorized switch from affecting STP.

```
show spanning-tree summary
show spanning-tree interface fastEthernet 0/1 detail
show errdisable recovery
show interfaces status err-disabled
```

Root Guard

Root Guard is used on ports where you expect the local switch to remain the designated side of the STP relationship. It prevents a downstream switch from becoming the root through that port.

Apply Root Guard on SW1 and SW2 ports facing SW3:

SW1:

```
configure terminal
interface gigabitEthernet 0/2
spanning-tree guard root
```

end

SW2:

```
configure terminal
interface gigabitEthernet 0/2
    spanning-tree guard root
end
```

Verify:

```
show spanning-tree inconsistentports
show spanning-tree vlan 10
show spanning-tree vlan 20
```

If a superior BPDU is received on a Root Guard-protected port, that port can become root-inconsistent and stop forwarding until the superior BPDU condition is removed.

Loop Guard

Loop Guard helps protect STP from a unidirectional failure where a port stops receiving BPDUs but remains physically up. It is particularly useful on redundant switch-to-switch links.

Option 1 - Enable Loop Guard globally on SW3

SW3:

```
configure terminal
spanning-tree loopguard default
end
```

Option 2 - Apply it to a specific redundant trunk port

```
configure terminal
interface gigabitEthernet 0/2
    spanning-tree guard loop
end
```

Verify:

```
show spanning-tree inconsistentports
show spanning-tree interface gigabitEthernet 0/2 detail
```

Important: Root Guard and Loop Guard serve different purposes. Root Guard protects the intended root/designated boundary from superior BPDUs; Loop Guard protects against loss of BPDUs on a non-edge STP path. Do not blindly apply every STP guard to every interface—choose the protection according to the port's role.

Verification command

```
show spanning-tree root
show spanning-tree
show spanning-tree root
show spanning-tree vlan 10
show spanning-tree summary
show spanning-tree inconsistentports
show interfaces trunk
show vlan brief
```

Verification Checklist

Command	Purpose
show spanning-tree root	See root bridge for each VLAN
show spanning-tree vlan 10	See root ID, bridge ID, port roles/states
show spanning-tree vlan 20	Verify VLAN 20 STP
show spanning-tree vlan 30	Verify VLAN 30 STP
show spanning-tree vlan 99	Verify management VLAN STP
show spanning-tree summary	STP mode and PortFast/BPDU Guard summary
show spanning-tree inconsistentports	Find Root Guard/Loop Guard inconsistencies
show interfaces trunk	Verify trunk links and allowed VLANs
show vlan brief	Verify VLAN/access-port assignment
show mac address-table dynamic	Verify MAC learning

